Comparison of Regression and Classification Models for Forecasting Thai Gold Prices

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***Abstract*— Accurate forecasting of gold prices is essential for investors, policymakers, and businesses due to gold's role as a financial asset and market hedge. This study compares classification and regression models to predict Thai domestic gold prices using daily price data and macroeconomic indicators. Classification models were applied to forecast short-term trends ("Up," "Down," "Neutral"), while regression models predicted the actual price. A time-series cross-validation approach with lagged and rolling features captured temporal dependencies in the data. Results indicate that Ridge Regression (alpha=1) achieved the lowest mean absolute error and highest R^2, demonstrating superior predictive performance for daily prices, while Logistic Regression achieved the highest F1-score for trend prediction. The findings highlight the effectiveness of regularised regression in modelling gold price dynamics and the challenges of trend classification due to market volatility and class imbalance.**

Keywords: ***Gold price forecasting, classification, regression, Time-series prediction, Feature engineering, Thailand***

I. Introduction

Gold is a widely recognised investment asset, valued for its liquidity, intrinsic worth, and role as a hedge against economic uncertainty and inflation[1],[2]. Its price is influenced by a combination of domestic and global macroeconomic factors, financial market trends, and investor sentiment[3]. Accurate forecasting of gold prices is essential for investors, financial institutions, and policymakers to manage risk and make informed decisions[4].

Recent studies have applied a range of models for gold price prediction, from traditional econometric approaches, such as ARIMA and Vector Autoregression (VAR)[4], [5], to machine learning techniques including Support Vector Machines, Random Forests, Gradient Boosting, and hybrid models that combine statistical and learning-based methods[6]. These approaches demonstrate improved performance over classical models by capturing non-linear patterns and high-dimensional dependencies. However, many existing studies focus on either regression or classification alone, use a limited set of macroeconomic indicators, and often do not fully exploit temporal features such as lagged values or rolling statistics. Furthermore, issues like multicollinearity, overfitting, and class imbalance in trend prediction are often insufficiently addressed[7], [8], [9].

This paper presents a complete framework for predicting Thai domestic gold prices utilising both regression and classification methodologies to solve these limitations. Regression techniques, including Linear, Ridge, and Lasso Regression, project daily price levels, whereas classification models, such as Logistic Regression, AdaBoost, and Gradient Boosting, anticipate short-term directional trends ("Up," "Down," or "Neutral"). Feature engineering methods, such as lagged and rolling statistics, are utilized to capture temporal dependencies, while time-series cross-validation is employed to maintain the sequential integrity of the data. The suggested framework seeks to establish a more reliable and precise system for predicting gold prices in Thailand, delivering insights for investors and a basis for subsequent study.

This work comprises five sections: Introduction; Related Works; Methodology; Results, and Conclusion.

II. Related Works

This section reviews the theoretical foundations and prior studies relevant to this research, encompassing the role of gold as an investment asset and the machine learning algorithms utilised for forecasting purposes.

A. Regression Models

1. Naïve Baseline

The Naive Baseline is a simple forecasting method that assumes the gold price at the next time step will equal the current price ($yt + 1 = \mathrm{yt}$) [10] . Despite its simplicity, it often performs well in financial time series due to strong autocorrelation and trend persistence. It serves as a benchmark for evaluating more complex regression models.

2. Linear Regression

Linear Regression characterizes the association between a dependent variable and one or more independent variables as a linear function. Linear Regression delineates the association between a dependent variable y and independent variables $x1, x2, \ldots, xp$ as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon \qquad (1)$$

In this equation, y represents the dependent variable (gold price), x_i denotes the predictor variables, β_i signifies the coefficients, and ϵ indicates the error term. It is extensively utilized because of its straightforwardness and clarity of interpretation [11], [12]. In financial research, it has been

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applied to forecast commodity prices, stock indices, and gold prices [13]. In this study, Linear Regression serves as a baseline to predict the daily gold price in Thailand using macroeconomic indicators, lagged prices, and rolling statistics.

3. Ridge Regression

Ridge Regression enhances Linear Regression by incorporating an *L2* penalty to mitigate overfitting and address multicollinearity among predictors.

$$\beta^{\wedge Ridge} = \min_{\beta} \left(\sum_{i=1}^{m} \left(y_i - \beta_0 - \sum_{j=1}^{n} \beta_j x_{ij} \right)^2 \right) + \lambda \sum_{j=1}^{n} \beta_j^2 \quad (2)$$

where λ is the regularisation parameter controlling coefficient shrinkage [14]. It has been employed in prior studies for predicting gold prices when macroeconomic indicators are highly correlated [15]. In this work, Ridge Regression is applied to stabilise coefficient estimates and improve the generalisation of gold price predictions. Thus, in this study, Ridge Regression is employed to enhance the robustness and predictive performance of gold price models using macroeconomic and autoregressive features.

4. Lasso Regression

Lasso Regression employs *L1* regularization, facilitating feature selection by reducing less informative coefficients to zero:

$$\beta^{\wedge Lasso} = \min_{\beta} \left(\sum_{i=1}^{m} \left(y_i - \beta_0 - \sum_{j=1}^{n} \beta_j x_{ij} \right)^2 \right) + \lambda \sum_{j=1}^{n} |\beta_j| \quad (3)$$

This property allows identification of the most important predictors [16]. Lasso has used for gold price prediction to identify key economic drivers while handling high-dimensional data [17]. In this study, Lasso is used to select the most influential features affecting Thai gold prices, such as lagged prices, exchange rates, and global gold prices.

B.Classification Models

1. Naïve Baseline (Classification)

In classification tasks, the Naive Baseline predicts the most frequent class found in the training dataset—for example, "Neutral" for gold price movement. This method provides a simple but important benchmark to determine if more complex models can demonstrate predictive improvement beyond merely guessing the majority class. Despite its simplicity, the Naive Baseline sets a performance floor and helps highlight challenges such as class imbalance in financial classification problems [18].

2. Logistic Regression

Logistic Regression is a simple yet powerful model for multi-class classification, widely applied in financial forecasting to predict directional price movements, credit risk, and default probability [19].

$$P(Y = c \mid X) = \frac{\exp(\beta_c^T X)}{\sum_{k=1}^{K} \exp(\beta_c^T X)} \quad (4)$$

where K is the number of classes, and βc are the class-specific coefficients. In this study, Logistic Regression predicts the trend of gold price movements (*Up*, *Down*, *Neutral*) using macroeconomic indicators and engineered features.

3. AdaBoost Classifier

AdaBoost is an ensemble technique that integrates several weak learners, usually decision stumps, to create a robust classifier. [20].

$$H(x) = \sin n(\textstyle\sum_{t=1}^{T} \alpha_t \, h_t(x)) \quad (5)$$

where $h_t(x)$is the weak learner at iteration t and α_t is its weight. It iteratively focuses on difficult-to-classify samples, improving predictive accuracy. Previous studies have applied AdaBoost to financial time series and commodity trend prediction [21].

4. Gradient Boosting

Gradient Boosting is an ensemble learning technique that constructs models in succession, with each subsequent model addressing the residual mistakes of its predecessor. The overarching prediction function at iteration m is:

$$\mathrm{F_m(x)} = \mathrm{F_{m-1}(x)} + \mathrm{v} \,.\, \mathrm{h_m(x)} \quad (6)$$

where $F_m(x)$ represents the boosted model at iteration m, $h_m(x)$ denotes the weak learner (often a decision tree), and v signifies the learning rate that regulates the influence of each model. Gradient Boosting can identify intricate nonlinear associations in financial data, rendering it suitable for predictive applications such as forecasting stock and commodities prices [22],[23]. In this study, Gradient Boosting was utilized for classification of Thai gold price directional movement, providing a useful benchmark against Logistic Regression and AdaBoost, despite challenges from class imbalance.

III. Methodology

This section presents the framework and methodology for the study including; Conceptual Framework, Data Collection, Data Preparation, Classification Modeling, Regression Modeling, and Performance Comparison.

A. Conceptual Framework

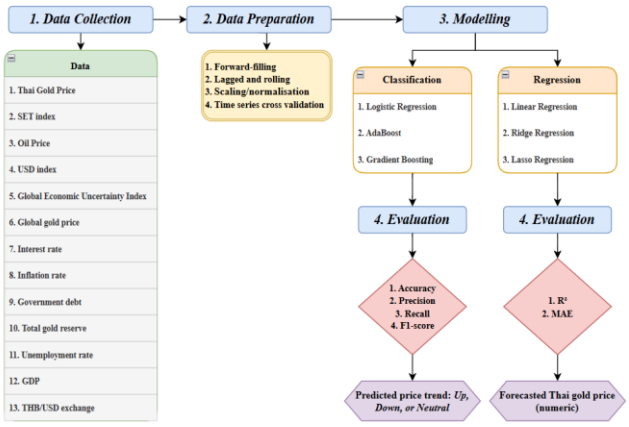

Fig. 1. Conceptual Framework of the Study

The study's conceptual framework outlines the methodology for predicting Thai domestic gold prices using both classification and regression approaches. Initially, daily gold prices and relevant macroeconomic indicators were collected and aligned to a daily frequency using forward filling. In the data preparation stage, missing values were addressed, non-numeric variables were converted, and lagged

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and rolling features were generated to capture temporal dependencies, followed by standardisation of all features. For modelling, classification models, including Logistic Regression, AdaBoost, and Gradient Boosting, were applied to predict short-term price trends, while regression models, including Linear, Ridge, and Lasso Regression, forecasted continuous gold prices, with hyperparameters tuned to improve predictive performance and reduce overfitting. The evaluation stage employed time-series cross-validation to maintain temporal integrity, using metrics such as accuracy, precision, recall, F1-score, mean absolute error, and R^2. This framework enables comparison of model performance, identification of influential features, and provides a foundation for future improvements, including advanced models or strategies to handle class imbalance.

B. Data Collection

The dataset utilised in this study comprises historical financial and macroeconomic indicators known to influence gold prices, spanning the period from 2019 to 2023. Data were collected from reputable public financial databases to ensure credibility and analytical reliability. This dataset includes daily records of the Thai domestic gold price, the Stock Exchange of Thailand (SET) index, oil prices, the United States Dollar Index (USD Index), inflation rates, interest rates, the USD exchange rate, the global economic uncertainty index, total gold reserve, Thailand's government debt, global gold prices, unemployment rates, and GDP figures[24]. The selection of variables was informed by economic theory and supported by prior research linking macroeconomic factors to commodity price dynamics. Indicators such as inflation and interest rates are known to affect gold prices via investor sentiment and perceived risk[25]. All data were temporally aligned and compiled into a single dataset for both regression and classification tasks.

C. Data Preparation

After collecting the datasets, preprocessing was performed to ensure consistency, completeness, and suitability for classification and regression modelling. Daily Thai domestic gold prices served as the target variable, with economic and financial indicators as features. Since feature frequencies varied (daily to annually), forward filling was applied to align them with daily gold prices, ensuring each observation had a complete set of explanatory variables[26].

All preprocessing was performed in Python using libraries such as pandas, numpy, and scikit-learn. Categorical features were converted to numeric formats, and missing values were handled using forward fill. For classification, gold price trends were encoded into three classes while regression modelling retained the continuous gold price as the target. Feature scaling was applied using z-score normalisation through StandardScaler, standardising all features to have zero mean and unit variance. This approach improved model convergence, prevented dominance of large-scale features, and enhanced comparability across different models[27], [28].

Autoregressive features, including one- and seven-day lags and seven-day rolling averages and standard deviations, were engineered to capture temporal dependencies in gold prices. The dataset was structured for time-series validation using walk-forward evaluation to preserve the temporal order and ensure accurate prediction[29]. These Python-based preprocessing steps transformed the dataset into a consistent format suitable for classification and regression modelling, enabling reliable performance evaluation.

D. Classification Modeling

The categorisation job sought to forecast short-term patterns in Thai domestic gold prices, categorised as "Up," "Down," or "Neutral." Time-dependent attributes were maintained by a five-fold time-series split (TimeSeriesSplit in Python), facilitating walk-forward validation and averting data leakage. Four models were utilized for classification: a Naive Baseline, Logistic Regression, AdaBoost Classifier, and Gradient Boosting Classifier. The Naive Baseline, executed using DummyClassifier, forecasts the predominant class in the training dataset. This basic model acted as a benchmark to assess the performance enhancement attained by more advanced algorithms. Logistic Regression was set with max_iter=1000 to guarantee convergence and random_state=1 for consistency. The model calculates class probabilities through a weighted amalgamation of features, elucidating the elements that predominantly affect gold price fluctuations. The AdaBoost Classifier was set with n_estimators equal to 100 and a learning_rate of 0.5. The number of estimators (100) was chosen to strike a compromise between model complexity and computing performance. A learning rate of 0.5 was selected to balance the influence of each weak learner, ensuring a trade-off between convergence velocity and stability. The Gradient Boosting Classifier was set with n_estimators=100, learning_rate=0.1, and max_depth=3. The number of estimators was established at 100 to ensure adequate learners for precise predictions while preserving computing efficiency. A learning rate of 0.1 was chosen to facilitate slow learning and mitigate overfitting, while max_depth=3 constrains the depth of each decision tree to diminish variance and improve generalization [30],[31]. This set of parameters is frequently advocated in financial time-series forecasting to achieve a compromise between model complexity and resilience. The integration of linear and ensemble approaches with optimized parameters facilitated both interpretability and the identification of non-linear dependencies. Consistent time-series validation guaranteed a reliable evaluation of model generalisability for predicting daily gold price trends [32].

E. Regression Modeling

The regression task aimed to predict the continuous value of daily Thai domestic gold prices. The preprocessed dataset incorporated lagged and rolling statistical features, such as one- and seven-day lags and seven-day rolling mean and standard deviation, to capture temporal dependencies alongside other explanatory variables. These features enabled the models to leverage historical dynamics in addition to broader market and macroeconomic factors. For modelling, four approaches were employed: Naive baseline, Linear Regression, Ridge Regression, and Lasso Regression. The Naive baseline predicted the previous day's gold price for each day, providing a simple reference for model performance. Linear Regression was included as a standard, interpretable model, serving as a benchmark to assess the benefit of regularisation and feature selection techniques. Ridge Regression and Lasso Regression were implemented with hyperparameter tuning using GridSearchCV. Ridge Regression controls the magnitude of model coefficients through the alpha parameter, tested over [0.01, 0.1, 1, 10, 100]. Smaller alpha values allow the model to closely fit the

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data, while larger values impose stronger regularisation to reduce overfitting. In the context of gold price prediction, this balance enables the model to capture meaningful relationships with market and macroeconomic indicators while maintaining stability against noisy fluctuations. Lasso Regression also uses alpha to perform both regularisation and feature selection, with tested values [0.001, 0.01, 0.1, 1, 10]. Lower alpha values retain more features, whereas higher values shrink irrelevant or weakly correlated predictors to zero[27], [33]. This characteristic is particularly useful for gold price forecasting, as it highlights the most influential economic and financial factors while reducing noise from less relevant variables, thereby improving interpretability and predictive robustness[34]. This combination of linear, regularised, and ensemble-inspired feature selection ensured robust prediction while mitigating multicollinearity and overfitting.

F. Performance Comparison

Evaluation was performed separately for classification and regression models, considering both accuracy and interpretability.

The objective of the classification models was to forecast the trajectory of gold prices. Accuracy quantified the overall ratio of right predictions, demonstrating the model's reliability in classification. Precision measures the ratio of accurate positive predictions, rendering it especially useful for evaluating the dependability of particular trend projections. Recall quantifies the ratio of accurately detected actual positives. The F1-score, a harmonic mean of precision and recall, offers a balanced assessment of classification ability, particularly advantageous in scenarios with imbalanced data. Weighted averages were utilized across categories (Up: 685, Down: 583, Neutral: 297) to rectify this disparity.[35]. For regression models, where the target was the continuous gold price, Mean Absolute Error (MAE) and the coefficient of determination (R^2) were used. The MAE measured the average amount of prediction errors in Thai Baht, providing a clear and interpretable measure of accuracy; lower MAE values signify that the model can accurately track daily price swings with minimum variance. R^2 evaluated the proportion of variance in real prices elucidated by the model, with elevated values indicating a greater capacity to encapsulate underlying patterns impacted by macroeconomic or market forces.[36]. Time-series validation with five-fold walk-forward splits preserved temporal order, ensuring realistic performance assessment. Baseline models provided reference points to quantify the added value of advanced techniques[37]. Using multiple metrics, baseline comparisons, and time-series validation, the evaluation framework provides a comprehensive assessment of predictive performance.

$$Accuracy = \frac{TP+TN}{TP+FP+FN+TN} \tag{7}$$

$$Precision = \frac{TP}{TP+FP} \tag{8}$$

$$Recall = \frac{TP}{TP+FN} \tag{9}$$

$$F1 = \frac{2\times Precision\times Recall}{Precision + Recall} \tag{10}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{11}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2} \tag{12}$$

IV. Results

This section presents the experimental findings of the proposed framework for predicting Thai gold investment using machine learning approaches. The results are organised into two parts: regression analysis for predicting the actual gold price in Thai Baht and classification analysis for forecasting gold price movements (Up, Down, Neutral).

A. Regression Results

Regression models were utilised to forecast Thai gold prices employing macroeconomic and financial data alongside lagged variables. The evaluation metrics included MAE and the R^2, averaged across five folds. Table I summarises the performance of regression model with different regularisation settings.

TABLE I. Regression Model Performance (Average across folds with lags)

Model	**MAE**	**R^2**
Naive	118.8	0.95
Ridge (α=1)	133.32	0.9674
Lasso (α=0.1)	153.15	0.8553
Ridge (α=0.01)	158.23	0.8523
Linear Regression	174.59	0.9251
Lasso (α=1)	275.54	0.9606
Lasso (α=10)	336.89	0.8847
Ridge (α=0.1)	378.64	0.9684

The results show that Ridge Regression consistently achieved strong performance, with α = 0.1 giving the best R^2 (0.9684), capturing nearly 97% of variance in Thai gold prices. Meanwhile, the Naive baseline also performed competitively due to strong autocorrelation in gold prices. Interestingly, Lasso Regression showed inconsistent behaviour: while α = 1 achieved a high R^2 (0.9606), other α settings (0.1, 10) reduced explanatory power significantly, suggesting that Lasso's feature selection penalisation can be sensitive to parameter tuning in this dataset. Overall, the regression results indicate that incorporating lagged prices and rolling statistics allows the models to capture short-term dynamics, while Ridge Regression balances accuracy and robustness against non-stationary and correlated features.

B. Classification Results

Classification model predict gold price trends (*Up*, *Down*, *Neutral*). Models evaluated include Naive Baseline, Logistic Regression, AdaBoost, and Gradient Boosting, with accuracy, precision, recall, and F1-score reported. Table 2 summarises the average results.

The results indicate that Naive Baseline achieved an accuracy of 0.4515, reflecting the high frequency of a dominant class in the dataset. Its strong performance relative to some ML models indicates the difficulty in predicting minority classes and highlights the impact of class imbalance in the dataset.

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Logistic Regression slightly outperformed other models in terms of F1-score (0.3287), indicating a better balance between precision and recall across classes. AdaBoost showed high recall for Up movements but struggled to correctly classify Neutral movements, often misclassifying them as Up or Down. Gradient Boosting achieved relatively higher precision but lower overall accuracy, reflecting challenges in generalizing across all classes.

TABLE I. CLASSIFICATION MODEL PERFORMANCE

Model	Accuracy	Precision	Recall	F1-score
Naive Baseline	0.4515	0.2055	0.4515	0.282
Logistic Regression	0.41	0.3282	0.41	0.3287
AdaBoost	0.4308	0.2636	0.4308	0.2959
Gradient Boosting	0.3338	0.3568	0.3338	0.2831

To further analyse the misclassification patterns, Fig. 2 shows the confusion matrix for AdaBoost. The classifier correctly identified most "Up" movements but struggled significantly with "Neutral", often misclassifying them as "Up" or "Down".

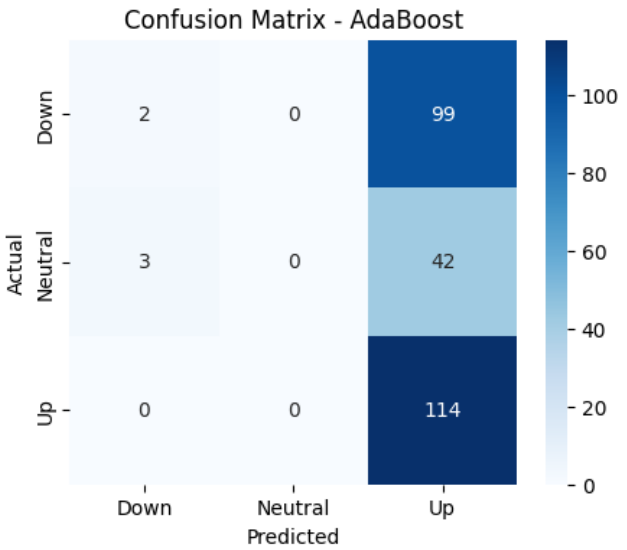

Fig. 2. CONFUSION MATRIX OF ADABOOST CLASSIFIER

C. Discussion of Findings

Overall, the results show that Ridge Regression is the most effective model for predicting actual gold prices, offering robust performance by handling multicollinearity in macroeconomic indicators. On the other hand, classification models remain challenging due to class imbalance and noise in the directional movement of gold prices. Nevertheless, Logistic Regression and AdaBoost provide reasonable baselines for directional prediction.

The feature analysis underscores the importance of global market indicators (e.g., global gold price, USD index, oil price) alongside domestic macroeconomic factors (e.g., SET index, Thai government debt) in shaping Thai gold prices.

V. CONCLUSIONS

This study compared classification and regression models for forecasting Thai domestic gold prices using market and macroeconomic indicators. Ridge Regression (α=1) achieved the best regression performance, with a MAE of 133.32 and R^2 of 0.9674, demonstrating high accuracy while mitigating overfitting. For classification, Logistic Regression obtained an F1-score of 0.3287 and accuracy of 0.41, reflecting moderate effectiveness in trend prediction. The study's strengths include the integration of both classification and regression approaches, the use of time-series cross-validation to preserve temporal integrity, and the implementation of lagged and rolling features to capture historical dependencies. However, limitations remain. Classification performance was constrained by class imbalance and market volatility, and regression models may not fully account for sudden exogenous shocks affecting gold prices. The forward-filling could introduce lag effects or bias. Furthermore, the study focused on linear and boosting-based models, potentially overlooking more complex non-linear techniques.

Future work may explore deep learning (e.g., LSTM, Transformer), address class imbalance, integrate real-time indicators, and apply interpretability techniques such as SHAP or LIME. In conclusion, Ridge Regression delivers strong predictive accuracy for price levels, while Logistic Regression offers modest trend classification. The study provides a basis for more advanced modelling in gold price forecasting.

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